Erlan Cassiano

Satellite Mechanical Engineer (Structures & Design)

Spacecraft Structures | Mechanical Design | FEA | Vibration Qualification & Test Correlation | MGSE

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PROFESSIONAL SUMMARY

Mechanical engineer with experience since 2014: Mechanical Architect and Space Systems Engineer at Visiona (since 2023) and propulsion trainee at Thales Alenia Space (2014–2016). Leads the mechanical and thermal architecture of SATVHR, a sub-300 kg Earth-observation satellite, with 4 junior engineers and 2 interns: mechanical design and configuration, structural analysis, handling equipment and the vibration qualification of its structural-thermal model (STM) in May 2026, with test–analysis correlation.

CORE COMPETENCIES

- **Structures & design:** spacecraft structures, sandwich panels, configuration, payload accommodation, mass budget, interface control documents (ICDs), deployables (VCUB1 solar-panel deployment test), lifting and handling mechanical ground support equipment (MGSE).
- **Analysis & CAD:** finite element analysis (FEA) with Femap/Nastran, thermal models (Thermal Desktop/SINDA), CAD (Creo Parametric, Onshape, SolidWorks), MATLAB, Python; drawings with geometric dimensioning and tolerancing (GD&T) and tolerance stack-ups to ISO GPS (geometrical product specifications).
- **Verification:** STM sine, random and sine-burst qualification (notching, modal survey) to tailored ECSS (European Cooperation for Space Standardization) standards and NASA GEVS (General Environmental Verification Standard) levels; test–analysis correlation; jitter measurement.
- **Delivery:** design for manufacturing, supplier follow-up (MGSE manufacture; thermal-controller specification through to acceptance), assembly procedures, non-conformance reviews.
- **Reviews:** preliminary requirements review (PRR), preliminary design review (PDR) and critical design review (CDR); test readiness review (TRR) and test review board (TRB) for environmental tests.

PROFESSIONAL EXPERIENCE

Visiona Tecnologia Espacial S.A. **Mar 2023 – Present**

Space Systems Engineer – Mechanical Architect *São José dos Campos, Brazil (hybrid)*

- Lead the mechanical and thermal architecture of SATVHR, a sub-300 kg, 5-year-life low Earth orbit (LEO) satellite in phase B/C; own its mass budget, trade-offs and ICDs.
- Sized and reviewed the sandwich panels (aluminium honeycomb; carbon-fibre facesheets for flight, aluminium on the STM); built, ran and reviewed Femap/Nastran and Thermal Desktop/SINDA models.
- Co-authored the STM test plan and specification and led its vibration qualification at INPE/LIT (Integration and Testing Laboratory), passed May 2026, correlating results with the FEM.
- Assembled and adjusted the laser set-up and measured jitter on the STM (main payload telescope mirror displacement, precision better than 0.1 μm), verifying reaction-wheel imbalance effects.
- Led the lifting and handling MGSE concept (built by a subcontracted supplier, tested during STM integration) and co-wrote the thermal-controller specification, following its supplier through to acceptance.
- Manage 4 junior engineers and 2 interns since Mar 2023 (trained 2); present mechanical and thermal at system reviews (PRR, PDR) towards the system, structure and thermal CDRs (end 2026).

Visiona Tecnologia Espacial S.A. **Jan 2023 – Mar 2023**

Space Systems Engineer (VCUB1 CubeSat) *São José dos Campos, Brazil*

- VCUB1 (12 kg CubeSat, operated in orbit): supported final integration, incl. the delta vibration and solar-panel deployment tests; revised the solar-panel assembly procedure and performed the final flight assembly.
- Ran the 6U deployer-rail fit-check, reviewed qualification test data and prepared integration tooling for the Falcon 9 launch (SpaceX Transporter-7 rideshare, Apr 2023).
- Co-authored the IAC 2024 paper on the VCUB1 thermal-vacuum test campaign (IAC-24-B4.IP.21), doing the data analysis and part of the writing.

Thales Alenia Space **Mar 2014 – Oct 2016**

Satellite Propulsion Engineer – SGDC Technology Transfer (on-the-job training) *Cannes, France*

- Showed that more than 16 kg of additional propellant could be loaded on Brazil's SGDC geostationary satellite, accounting for tank deformation under pressurisation, adding ~2.5 months of service life (~R$36M).
- Worked across the bipropellant propulsion lifecycle (tanks, valves, lines, interfaces), from requirements specification and design to production, procedures, testing and operation.
- Witnessed, as a trainee, all propulsion-related SGDC tests and selected alignment and thermal tests during the satellite's test campaign in Cannes.

OTHER EXPERIENCE

Junior Space Project Analyst, Funcate (Brazilian space-science foundation) Feb 2020 – Nov 2020, Brasília
Brazilian Space Agency (AEB): model-rocketry launch documentation; Alcântara Launch Centre authorisations.

RuaDois: Project Coordinator, Head of Product, COO, Technical Consultant Oct 2018 – Jul 2024, São Paulo
Team grew to 20 staff + 50+ inspectors and visit companions; 20,000+ properties and 6,000+ inspections in 1.5 years.

SET-G Fellow, CNPq (Brazil's research council): SINDAE satellite project Oct 2016 – Sep 2017, Brasília

Co-founder & CTO, Fox17: cement-industry filters and separators Jul 2015 – Jun 2023, Brasília
Designed 80 t, 100 m+ steel chimneys and their flanges for export.

Project Designer, Autotrac: SolidWorks test-rig structures; CNC prototypes Feb 2014 – Apr 2014, Brasília

EDUCATION

Instituto Tecnológico de Aeronáutica (ITA), São José dos Campos, Brazil **Jul 2025 – Present**
M.Sc. in progress (expected Jul 2027), EAM-1 – Aeronautical Design, Structures, and Aerospace Systems
Thesis: *Development of a Magnetohydrodynamic (MHD) Magnetic-Fluid Reaction Wheel for CubeSat Attitude Control*; advisor: Prof. Tiago de Paula.

Universidade Cruzeiro do Sul, Online **Jun 2019 – Nov 2021**
MBA in Data Analysis, BI and Big Data (450 h)

Universidade de Brasília (UnB), Brasília, Brazil **Jun 2008 – Jun 2013**
Bachelor's in Mechanical Engineering
Research volunteer, hybrid rocket static-fire test benches (Mar 2012 – Jun 2013); metrology and reverse-engineering intern (Jan 2012 – Mar 2013). Extracurricular: Apuama Racing (Formula SAE), N-Prize.

PUBLICATIONS

- Silva, S. et al., incl. **Cassiano, E.** (2024). *Vcub1 Brazilian CubeSat Thermal Vacuum Tests Campaign.* 75th International Astronautical Congress (IAC 2024), Milan, IAC-24-B4.IP.21. doi:10.52202/078365-0126.
- Bertoldi, A. E. M.; **Cassiano, E. R. S.** et al. (2016). *A Study of the Atomization Process Using Numerical Simulation.* CILAMCE 2016, RIPE (UnB) v.2 n.31. CFD (ANSYS CFX) of a pressure-swirl atomiser.

CERTIFICATIONS & COURSES

Designing Space Missions and Systems (Teaching Science & Technology Inc., 2024, 40 h) • CNC machining-centre programming, setup and operation for aerospace (SENAI, Brazilian industrial training service, 2024, 120 h); CAM programming (SENAI) • Creo Parametric (Mult-e, 2023, 20 h) • Space Mission Structures (Applied Technology Institute, 2023, 40 h)

HONOURS & AWARDS

- AEB recognition, Mar 2017: propellant-loading method that extends the lifetime of the SGDC satellite.
- FAP-DF (Federal District Research Support Foundation) funding, Sep 2019: Omega electromechanical battery for photovoltaic systems (technical lead, Fox17).

VOLUNTEERING

COBRUF (Brazilian university rocketry competition): Launch Base Subsystem Lead, design of a mobile launch platform (Oct 2013 – Jan 2016) • N-Prize team (UnB): co-founder, systems engineering and hybrid rocket motor (Jun 2010 – Jul 2012)

LANGUAGES

Portuguese: native • English: professional working proficiency (Duolingo English Test 129, Aug 2026) • French: limited working proficiency